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|---|---|---|---|
| **3**<br>**Algorithms on graphs II** | 3.1 | **Algorithm for finding the shortest route around a network, travelling along every edge at least once and ending at the start vertex (The Route Inspection Algorithm).** | **Also known as the 'Chinese postman' problem.**<br>**Students will be expected to use inspection to consider all possible pairings of odd nodes. The network will contain at most four odd nodes.**<br>If the network has more than four odd nodes then additional information will be provided that will restrict the number of pairings that will need to be considered. |

*This paper is also the **Paper 4 option 4D** paper and will have the title '*Paper 4, Option 4D: Decision Mathematics 1'. Appendix 9*, '*Entry codes for optional routes'* shows how each optional route incorporates the optional papers.

| Topics | What students need to learn: | | |
|---|---|---|---|
| | Content | | Guidance |
| **3**<br>**Algorithms on graphs II**<br>*continued* | 3.2 | The practical and classical Travelling Salesman problems.<br>The classical problem for complete graphs satisfying the triangle inequality. | The use of short cuts to improve upper bound is included. |
| | | Determination of upper and lower bounds using minimum spanning tree methods. | The conversion of a network into a complete network of shortest 'distances' is included. |
| | | The nearest neighbour algorithm. | |